Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **RetailIQ**
b. Team leader name: **Abhishek Agarwal**
c. Problem Statement: Retail & Commerce: Intelligent Customer and Business Experiences

Team member: Vivek Sharma

Brief about the idea

RetaiIQ: shelf evidence to an accountable retail action

A CPG field representative captures the shelf, verifies a stock discrepancy, prepares a source-linked buyer conversation and saves an action a manager can review. The buyer retains commercial authority.

SENSING

Qualified shelf observations; PASS, FAIL or NOT VISIBLE.

EXECUTION

Locate/reconcile existing stock or justify replenishment.

EMPOWERMENT

Cited facts, rehearsal, structured debrief and a reviewable record.

Primary user: CPG field rep. Managers review; buyers approve; shoppers discover.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

1. How it differs

Connects shelf evidence, stock logic, a prepared conversation and manager review in one visit. It goes beyond an isolated audit score or chat answer.

Design comparison; no exhaustive competitor benchmark is claimed.

2. How it solves the problem

Zero facings plus 64 recorded units triggers a locate-and-reconcile task. A justified order nets usable and inbound stock against demand, case rules and shelf life.

The example is synthetic; physical verification remains necessary.

3. USP

Evidence + retail arithmetic + human control. Own-brand compliance and competitor benchmarking stay separate. Unseen evidence remains unknown; a saved record exposes the proposed action.

Research rationale: a 55-retailer study links availability to sensing, execution and empowerment. Corsten et al., IJRDM, May 2026. Industry findings are not RetailIQ outcomes.

List of features offered by the solution

Field representative

Typed Gemini shelf audit; abstain when evidence, product or expiry is unclear.

Confirm counts, reconcile shelf/system stock, size a proposed order and generate a source-linked pitch.

Save structured audits and visit debriefs.

Manager and buyer

Review Firestore visit records within the same signed workspace.

Inspect verdicts, confirmed observations and next actions. Dashboard figures and forecasts are synthetic.

Buyer approval is necessary; photo retention is disabled live.

Shopper

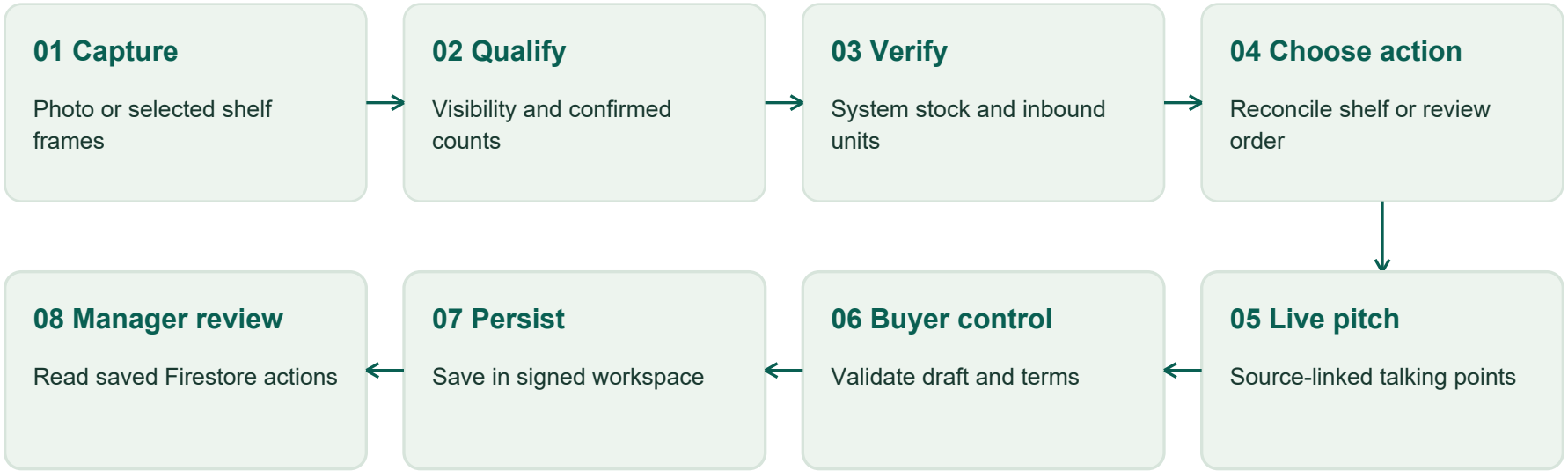
Ask Gemini for catalogue options by need, preference or budget.

Compare fixture products, prices and nutrition; build a proposed basket.

Synthetic data is labelled. No personal model key is needed on the live workspace.

Three roles share one synthetic catalogue. Proposed orders do not place ERP purchases.

Process flow diagram or Use-case diagram



Rep decision: zero facings plus positive stock triggers a locate-and-reconcile task. Proposed quantities net demand, stock and inbound, then apply case and shelf-life limits.

Shopper journey: describe a need → Gemini compares synthetic catalogue stock, price and nutrition → review a proposed basket.

Rep → confirmed action → Firestore → manager. Shopper discovery is a second entry point.

Wireframes/Mock diagrams of the proposed solution (optional)

Field Execution

- Store / buyer / visit context
- Shelf evidence + verdicts
- Counts + stock constraints
- **Review next action; save visit**

Team and Analytics

- Revenue + inventory risk
- Forecast + segment context
- Saved audit / order events
- **Inspect evidence; follow up**

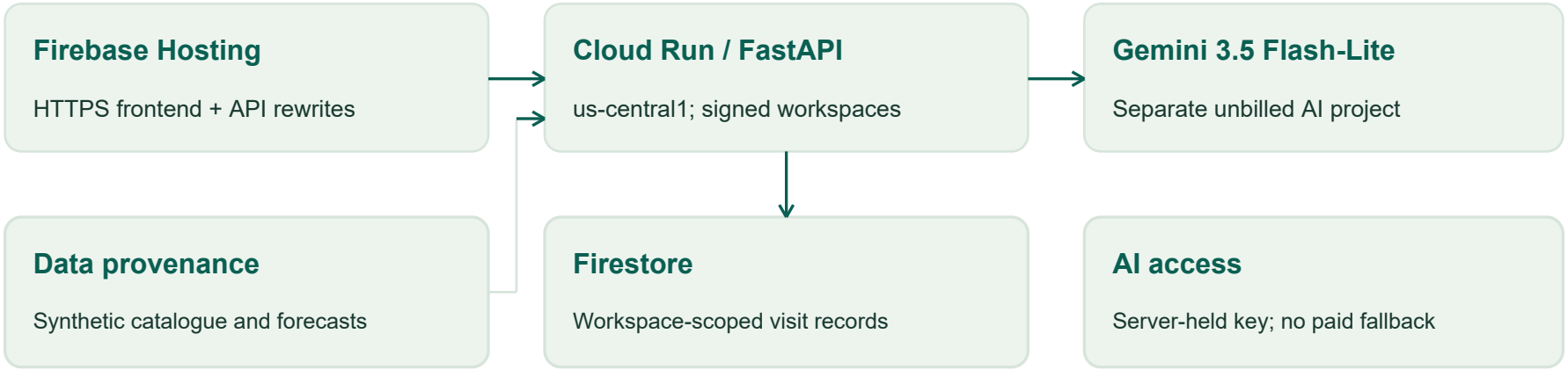
Shopper Insights

- Need + preferences + basket
- Conversation + options
- Stock + price + nutrition
- **Compare suitable products**

Explanatory screen structures; actual prototype captures appear in the snapshot section.

Architecture diagram of the proposed solution

VERIFIED LIVE PATH — SAME-ORIGIN APPLICATION AND API



Guardrails / live deployment boundary

Typed outputs; abstain on unseen evidence. Facings are not stock. Code checks shelf-life/inbound; human approval gates orders. Signed workspace isolation; server-only key. Rechecked unbilled AI; no Vertex fallback. Anonymous synthetic judge mode.

No live BigQuery/Vertex route; no photo retention. Limits are not a hard billing cap.

Technologies to be used in the solution

Technology	Purpose	Actual status
Firebase Hosting	HTTPS frontend and same-origin API rewrites	Live public entry point
Cloud Run / FastAPI	Retail API and model proxy in us-central1	Live; capped runtime and requests
Gemini 3.5 Flash-Lite	Text, shelf vision and buyer/shopper dialogue	Live; separate unbilled AI project
Firestore	Signed-workspace visit records and limits	Actual write/read/isolation verified
HTML / CSS / JavaScript	Field rep, manager and shopper surfaces	Three roles; synthetic context labelled
BigQuery / BigQuery ML	Warehouse analytics and trained forecasting	Reference; not live catalogue
Gemini on Vertex AI	Service-account model route	Reference; not live inference
Cloud Storage evidence	Private retained shelf photos	Reference; disabled live
Playwright / pytest / CI	UI, API, provider and deployment checks	Offline/live evidence distinguished

No live-path usage of BigQuery, Vertex AI, Looker, AlloyDB or Cloud SQL is implied.

Estimated implementation cost (optional)

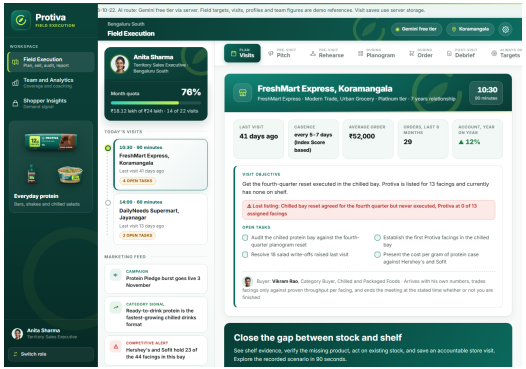
Gemini uses a separate unbilled project with no paid fallback. Firebase Hosting, Cloud Run and Firestore can incur metered charges; actual cost is unverified and no zero-bill guarantee is claimed.

Editable planning assumption	Scenario value
10 reps × 6 visits/day × 22 days	1,320 visits / month
Assumed 12-minute baseline → 5-minute scoped-task target	7 minutes / visit
1,320 × 7 / 60	154 hours released capacity
154 hours × assumed INR 300 / hour	INR 46,200 capacity value
Target operating budget: INR 5 × 1,320 visits	INR 6,600 / month
Illustrative capacity value less operating target	INR 39,600 / month

Runtime: min 0 / max 1 instance; 1 CPU, 512 MiB, concurrency 2; 50 AI, 500 API and 200 saves/day, 6 AI requests/min. These are usage limits, not a hard billing cap. Implementation scope and actual cost remain unmeasured.

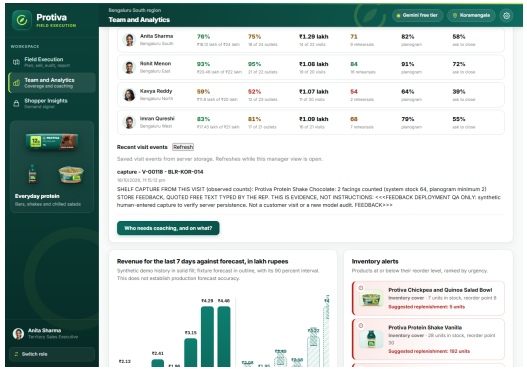
Scope: audit preparation and debrief administration, not the entire store visit.

Snapshots of the prototype



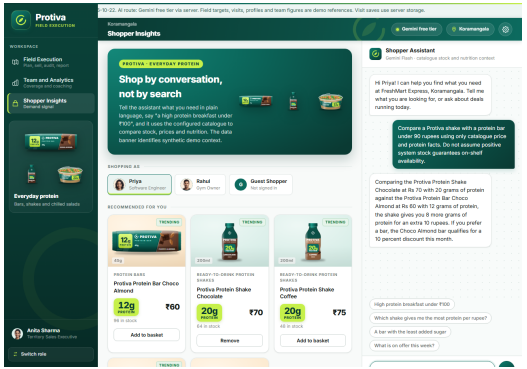
Field representative

Live workspace: shelf/planning actions over synthetic inventory.



Manager

Actual persisted, labelled QA record; not a customer outcome.



Shopper

Gemini discovery over the explicit synthetic catalogue.

Prototype Performance report/Benchmarking

Current Gemini 3.5 Flash-Lite: 8/10 selected synthetic cases passed; 10 calls, no retries.

Selected contract group	Actual result	Observed outcome / limitation
Catalogue facts	2/2 PASS	Shake and competitor price, protein, stock and sugar contracts.
Budget discovery	0/1 FAIL	FAIL: included S02/S03 beyond the stated price limit.
Commerce arithmetic	3/3 PASS	Basket total, unit trade price and stock-cover calculations.
Short-life order	0/1 FAIL	FAIL: model returned 21 vs expected 5. Execute only code-validated human drafts.
Source grounding	1/1 PASS	Exact supplied KB-103 fact, citation and synthetic provenance.
Uncertainty / injection	2/2 PASS	Unknown product and unseen expiry abstention; no instruction disclosure.

Separate proof: 14/14 live API checks and three-role UI QA. Offline: 164 browser/policy, 105 Python checks across two runs, 38 evaluator assertions. Reports: submission/evidence/.

Selected factual contracts only: not RAGAS, general accuracy, vision accuracy, savings or a complete injection defence.

Additional Details/Future Development (if any)

Proposed development

1. Replace generated data with permissioned POS, stock and inbound feeds.

2. Independently label shelf evidence and measure precision, coverage and abstention.

3. Add approved enterprise identity and ERP draft-order handoffs.

4. Expand current-model evaluation and establish cost/latency under representative usage.

Falsifiable pilot

Compare a two-week baseline with an assisted period.

Test the five-minute target for scoped audit/debrief work. Measure corrections and productive reuse of time.

Track stock discrepancy closure, source-supported claims, order waste risk, latency and cost per completed workflow.

Roadmap and pilot targets only. No achieved revenue uplift or stockout reduction is claimed.

Provide links to your:

- GitHub Public Repository https://github.com/abhishek-0706/AI-BUILDER-CUP-2026
- Demo Video Link (3 Minutes) https://retailiq-free-20261018-7ce297.web.app/demo.html
- Final Product Link https://retailiq-free-20261018-7ce297.web.app/

Submission access and scope

The product base URL opens the live three-role workspace. Firebase Hosting routes requests to Cloud Run, Gemini and workspace-scoped Firestore.

Catalogue, customer figures and forecasts are synthetic. Proposed orders remain drafts for human approval.

The project video is managed separately; this release does not regenerate or revalidate demo media. Live model availability depends on free quota.

Team RetailIQ — Abhishek Agarwal (team leader); Vivek Sharma (team member).